

Heat transfer part 2

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Plan for today

1. Questions or feedback
2. Class exercise on five systems
3. Continue work on problems 1 & 2, group check-in

- Heat transfer constitutive equations
- Fourier's law, Newton's law, advection equations
- Conservation and linking equations
- Parameters: specific heat capacity, heats of fusion and vaporization
- Analytical versus numerical models
- Python modules
- Python in VS Code

In groups, read the description of the system, and in 5 minutes come up with:

1. conservation equation (accumulation = ...),
2. constitutive equation (Fourier's law, ...),
3. parameters (specific heat capacity, ...),
4. dimensions (0D/lumped parameter, 1D, ...), and
5. simplifications (linear gradient, ...).

Then present your answers (share screen, use mic or chat).

Enter group name and members in the chat

1. Swimming pool

Nighttime heat loss from a swimming pool with a floating cover for insulation and possible external heater with some maximum heat output (heat pump, but the source does not matter).

2. Thermal shield

Maximum temperature at the base of a spacecraft thermal shield exposed to a fixed surface temperature for a fixed amount of time. (Hint: Are a linear gradient or lumped parameter appropriate here?)

3. Electric kettle

Electricity required to get water in a kettle to boil and the impact of kettle insulation and delay in shutting off after boiling point reached.

4. Solar oven

Steady-state temperature of a solar oven. (Hint: Have we covered the relevant modes of heat transfer?)

5. Insect agriculture

Heat removal from an insect rearing building by ventilation.